

DeepSeek-V2: Efficient MoE Language Model with Multi-

A Strong, Economical, and Efficient Open-Source LLM

236B

Total Parameters

21

Activated / Token

B

128

Context Length

K

8.1T

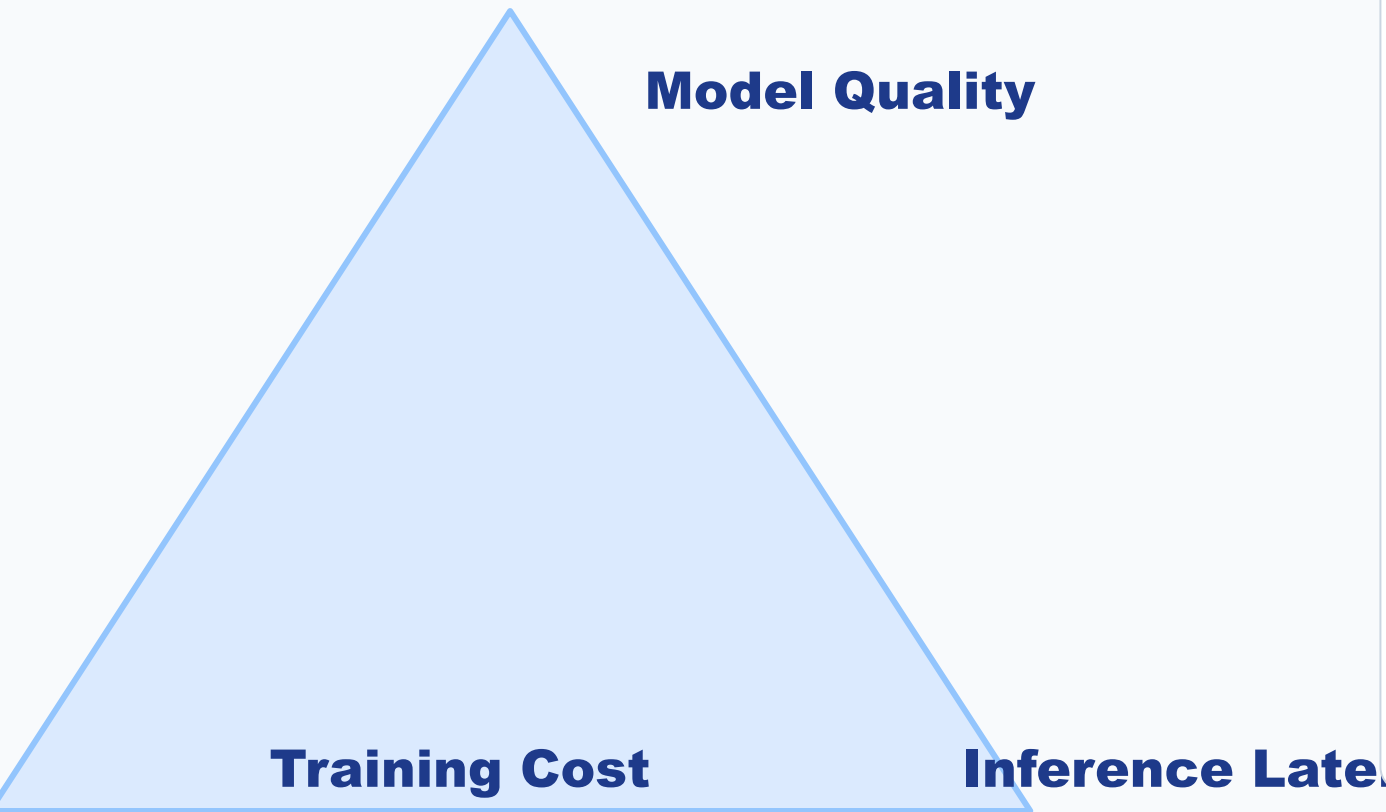
Pretraining Tokens

Post-training pipeline

- Supervised Fine-Tuning (SFT): 1.5M conversational sessions
- Reinforcement Learning: Group Relative Policy Optimization (GRPO)
- Objective: maximize capability while minimizing active compute and memory

LLM scaling creates a three-way tradeoff between quality, cost, and latency

Takeaway: Dense scaling alone is unsustainable; efficient architecture is the key lever.



Why conventional scaling breaks

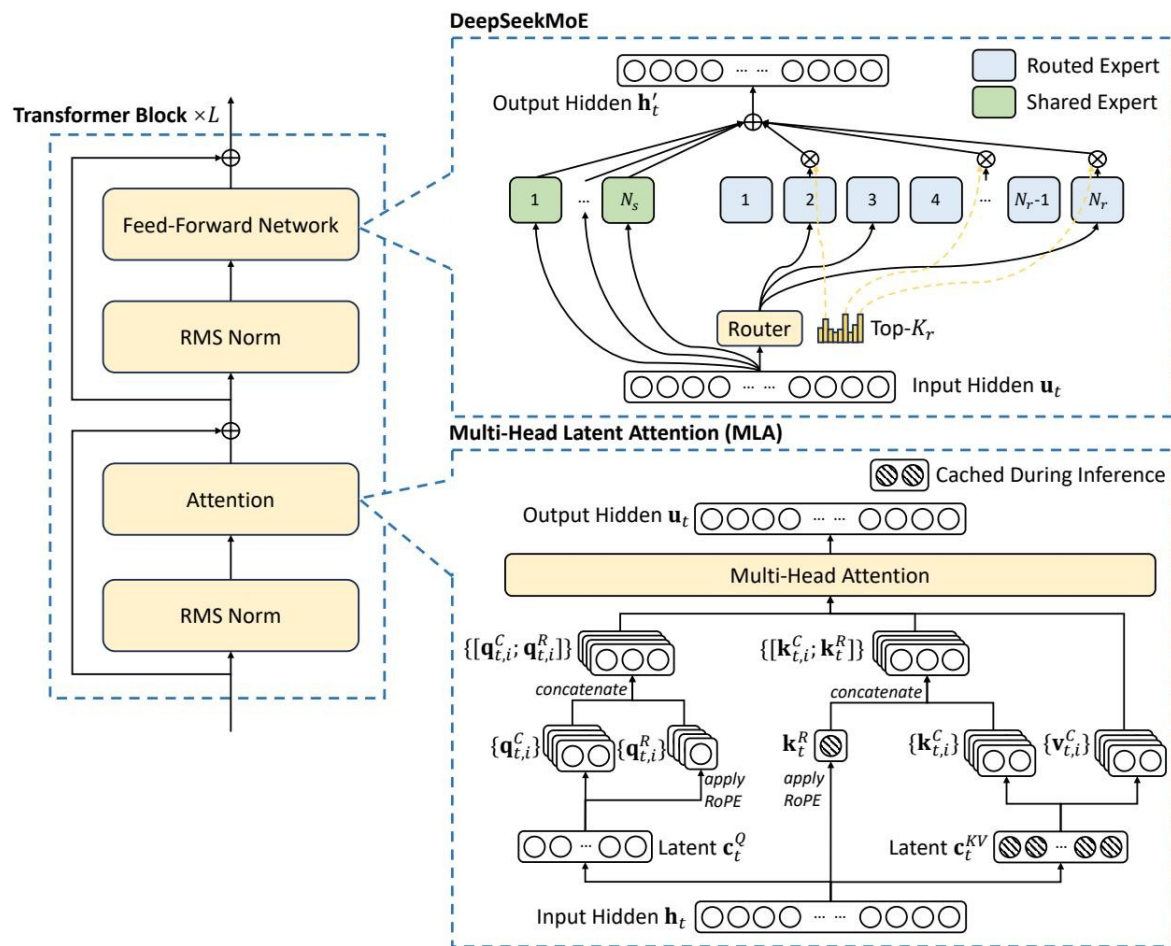
- Larger dense models improve intelligence but raise cost
- Autoregressive decoding grows KV cache with sequence length
- GQA/MQA cut cache, but can reduce quality relative to full attention
- Need sparse activation + cache-efficient attention w/ large context

DeepSeek-V2 hypothesis:

MLA + DeepSeekMoE can push the Pareto frontier on cost/latency

DeepSeek-V2 combines MLA attention and DeepSeekMoE FFN

Takeaway: Both attention and FFN pathways are redesigned for efficiency.



Architecture implications

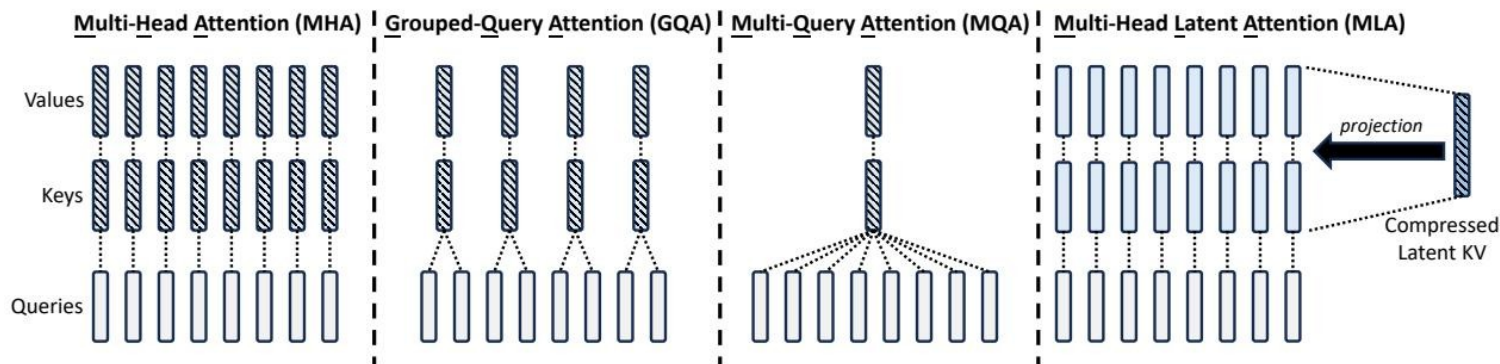
- MLA replaces standard MHA
- DeepSeekMoE replaces dense FFN
- 236B total, only 21B active
- Supports 128K context window

Design objective

Keep model quality at the 70B-class level while lowering active compute and memory.

MLA compresses KV states into latent vectors, then reconstructs

Takeaway: MLA outperforms MHA while reducing KV cache by 93.3%.



Mechanism summary

- 1) Cache compact latent vector c_t^{KV} instead of full K/V
- 2) Reconstruct keys, values, and decoupled RoPE positional key
- 3) Maintain quality while lowering inference memory pressure

93.3%

KV cache reduction vs DeepSeek 67

DeepSeekMoE isolates shared knowledge and scales specialization

Takeaway: Sparse expert activation preserves capability while reducing active compute.

Expert structure

Shared experts (N_s): always activated

Routed experts (N_r): top- K_r per token

- Fine-grained segmentation increases specialization
- Shared/routed split avoids duplicated common skills

Training efficiency controls

- Device-limited routing ($\leq M$ devices/token)
- Auxiliary losses for expert/device load balance
- Token-dropping during training prevents overflow
- No inference-time quality penalty from dropping

Resulting operating point

236B total parameters with only 21B activated per token.

Versus DeepSeek 67B, DeepSeek-V2 shifts the efficiency frontier

Takeaway: Major gains in training cost, memory footprint, and generation speed.

DeepSeek 67B

Training cost baseline

KV cache baseline

Generation throughput baseline

DeepSeek-V2

↓ **42.5%** training cost

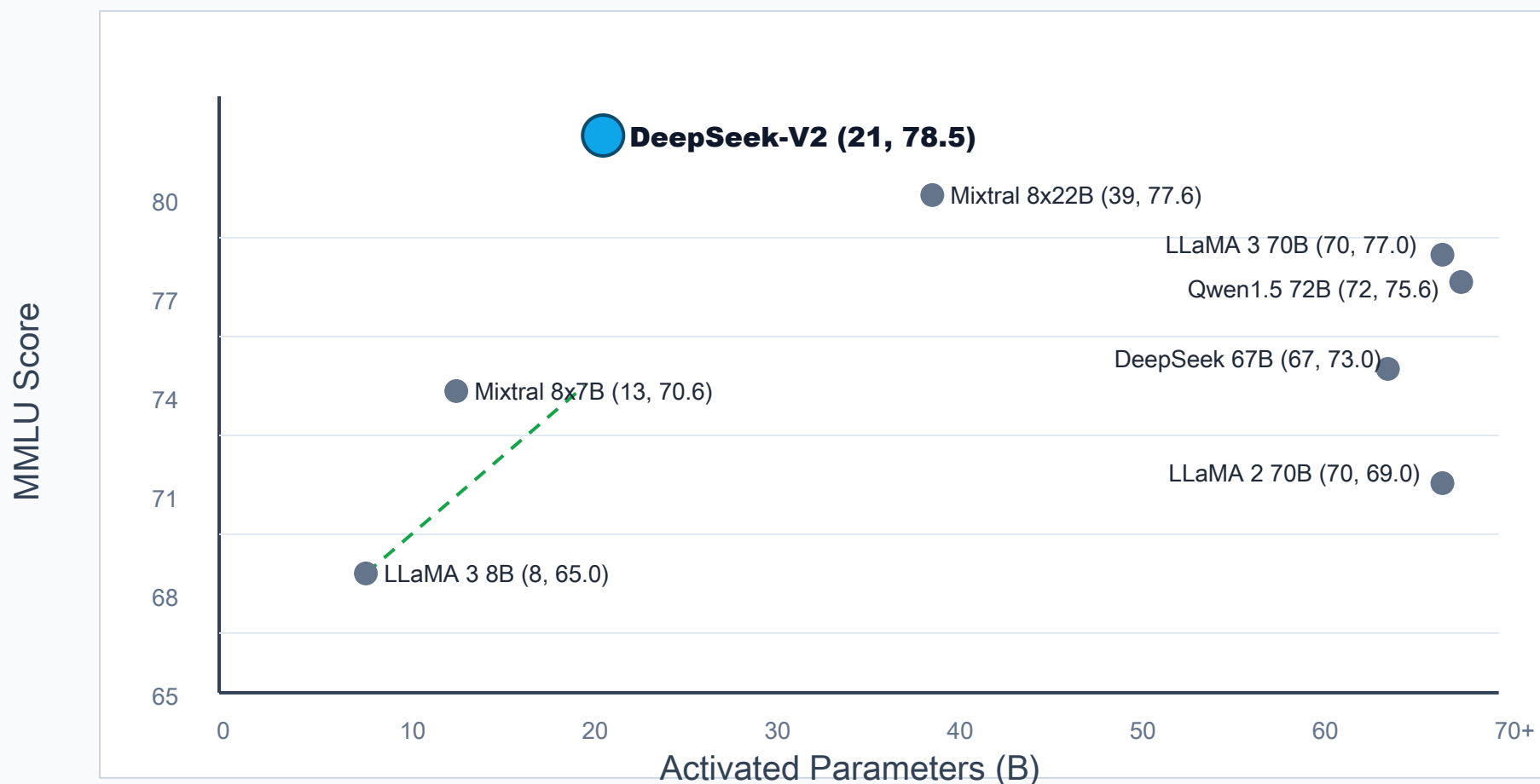
↓ **93.3%** KV cache size

↑ **5.76×** max generation throughput

Relative metrics only, as reported in source brief.

DeepSeek-V2 is Pareto-optimal on MMLU versus activated parameters

Takeaway: DeepSeek-V2 delivers the highest MMLU among models with far fewer active parameters.



Interpretation

- DeepSeek-V2 sits above higher-activation baselines
- 21B active parameters outperform 70B-class peers
- Indicates strong capability per active parameter

Post-training alignment yields state-of-the-art open-source chat qu

Takeaway: SFT + GRPO translates architectural efficiency into strong chat behavior.

DeepSeek-V2 Chat (RL) benchmark outcomes

AlpacaEval 2.0 (LC win rate) **38.9%**

MT-Bench (overall) **8.97**

AlignBench (overall, Chinese) **7.91**

Reported positioning:
strongest open-source Chinese chat model; competitive English quality

Process link

- 1) Efficient pretraining (8.1T)
- 2) SFT on 1.5M sessions
- 3) GRPO RL refinement

=> alignment quality without
giving up efficiency gains

DeepSeek-V2 demonstrates a practical Pareto path for open

Bottom line: high-end quality can be achieved with lower active compute and memory.

Key takeaways

- MLA: better-than-MHA quality trend with 93.3% KV cache reduction
- DeepSeekMoE: 236B total scale with only 21B activated parameters
- Relative to DeepSeek 67B: 42.5% lower training cost, 5.76× higher throughput
- MMLU 78.5 at 21B active params outperforms several 70B-class baselines
- Alignment results indicate strong open-source multilingual chat competitiveness

Implication: DeepSeek-V2 is positioned on a superior cost-quality-latency frontier.